

Lab 2 – Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #44 – Database Query Performance Profiler

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1. Epics

The 5 functional requirements from Lab 1 were grouped into 5 Epics in Jira. Each Epic represents a major feature area of the Database Query Performance Profiler.

Epic ID	Epic Name	Description	Related FR
DBQP-1	Log Ingestion	Covers all functionality for getting slow query log data into the system, whether via file upload or live agent streaming.	FR-002
DBQP-2	EXPLAIN Plan Analysis	Covers parsing of Postgres and MySQL EXPLAIN plans and detecting full table scans on large tables.	FR-001
DBQP-3	Index Recommendations	Covers the logic to scan query patterns and identify which columns are missing indexes.	FR-003
DBQP-4	Weekly Digest	Covers the automated Monday email report with the top 10 slow queries, recommendations, and week-over-week trend.	FR-004
DBQP-5	Dashboard	Covers the live real-time dashboard showing active slow queries, their plans and fixes, with database/severity/time filters.	FR-005

2. User Stories & Story Points

User stories were written for each Epic using the format: As a [role], I want [goal], so that [benefit]. Story points were assigned using the Fibonacci scale (1, 2, 3, 5, 8, 13) based on complexity and effort.

Story ID	User Story	Epic	Priority	Story Points
DBQP-6	As a DBA, I want to upload slow query log files so that I can start analysing past performance.	Log Ingestion	High	3
DBQP-7	As a DBA, I want logs streamed live from a database agent so that I get real-time query data without manual uploads.	Log Ingestion	High	5
DBQP-8	As a DBA, I want to upload an EXPLAIN plan so that I can see what the query is actually doing internally.	EXPLAIN Plan Analysis	High	3
DBQP-9	As a DBA, I want sequential scan warnings flagged automatically so that I know which queries need fixing without reading every plan myself.	EXPLAIN Plan Analysis	High	5
DBQP-10	As a DBA, I want index recommendations shown alongside each slow query so that I know exactly what to add to fix it.	Index Recommendations	High	5
DBQP-11	As a DBA, I want to see which columns appear most in WHERE/JOIN/ORDER BY so that I can prioritise which indexes to create first.	Index Recommendations	Medium	3
DBQP-12	As a Backend Lead, I want a weekly email every Monday with the top 10 slowest queries so that I can track what's degrading performance over time.	Weekly Digest	Medium	5
DBQP-13	As a Backend Lead, I want a week-over-week trend comparison in the digest so that I can see if things are improving or getting worse.	Weekly Digest	Medium	3

Story ID	User Story	Epic	Priority	Story Points
DBQP-14	As a DBA, I want a live dashboard of slow queries happening right now so that I can act on issues during peak load.	Dashboard	Medium	5
DBQP-15	As a DBA, I want to filter the dashboard by database name and severity so that I can focus on the most critical queries.	Dashboard	Low	2

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3. Jira Backlog – Epics and User Stories

The screenshot below shows the Jira backlog after creating all 5 Epics and 10 User Stories. Sprint 1 was populated with the 4 highest-priority stories (16 story points total — DBQP-6: 3pts, DBQP-7: 5pts, DBQP-8: 3pts, DBQP-9: 5pts).

DB Query Profiler
Software project

Roadmap
Backlog
Active sprints
Reports

Projects / DB Query Profiler / Backlog

Backlog

EPICS

- Log Ingestion 2
- EXPLAIN Plan Analysis 2
- Index Recommendations 2
- Weekly Digest 2
- Dashboard 2

Sprint 1 Sep 3 – Sep 10 16 [Start sprint](#)

☐	DBQP-6	As a DBA, I want to upload slow query log files so that I can start analysing past performance.	LOG INGESTION	3
☐	DBQP-7	As a DBA, I want logs streamed live from a database agent so that I get real-time query data.	LOG INGESTION	5
☐	DBQP-8	As a DBA, I want to upload an EXPLAIN plan so that I can see what the query is doing.	EXPLAIN PLAN ANALYSIS	3
☐	DBQP-9	As a DBA, I want seq scan warnings flagged automatically so that I know which queries need fixing.	EXPLAIN PLAN ANALYSIS	5

Backlog 6 issues

☐	DBQP-10	As a DBA, I want index recommendations shown alongside each slow query...	INDEX RECOMMENDATIONS	5
☐	DBQP-11	As a DBA, I want to see which columns appear most in WHERE/JOIN/ORDER BY...	INDEX RECOMMENDATIONS	3
☐	DBQP-12	As a Backend Lead, I want a weekly email every Monday with the top 10 slowest queries...	WEEKLY DIGEST	5
☐	DBQP-13	As a Backend Lead, I want a week-over-week trend comparison in the digest...	WEEKLY DIGEST	3
☐	DBQP-14	As a DBA, I want a live dashboard of slow queries happening right now...	DASHBOARD	5
☐	DBQP-15	As a DBA, I want to filter the dashboard by database name and severity...	DASHBOARD	2

Figure 1 – Jira Backlog view showing Epics panel and Sprint 1 with story point assignments

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4. Sprint 1 – Active Sprint Board (Sep 3 – Sep 10)

Sprint 1 ran from September 3 to September 10. It included 4 stories from the Log Ingestion and EXPLAIN Plan Analysis epics — 16 story points in total. Sep 3 was the sprint start and planning day; no tasks were moved to Done yet. Work began on Sep 4. The board below was captured mid-sprint with DBQP-6 and DBQP-8 already in Done and DBQP-7 and DBQP-9 in progress.

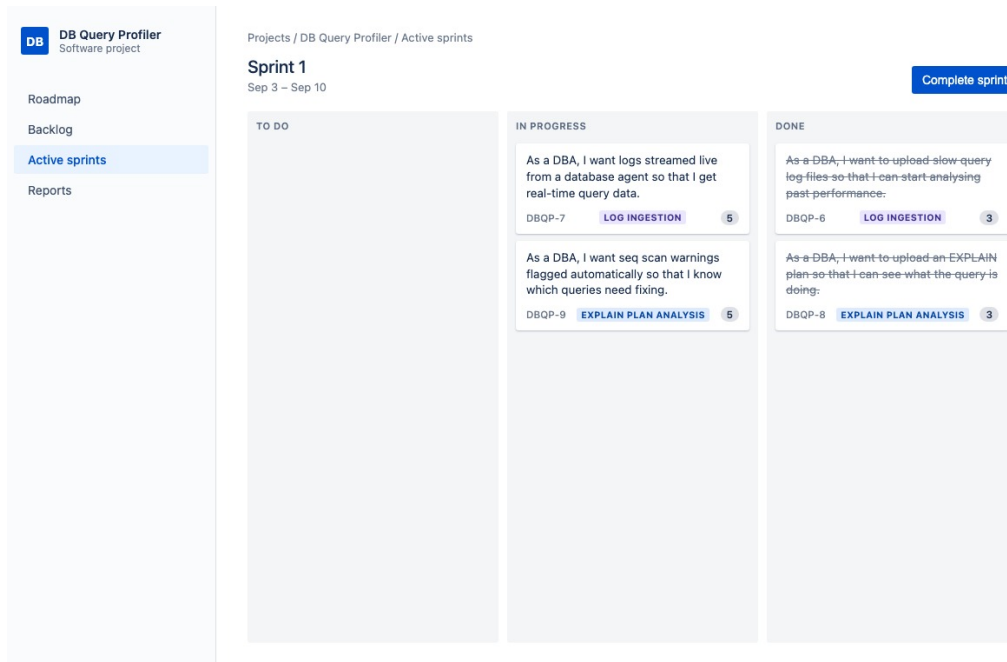


Figure 2 – Sprint 1 active sprint board mid-sprint

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5. Sprint 1 – Burndown Chart

The burndown chart for Sprint 1 shows the sprint starting on Sep 3 with 16 story points. Sep 3 was a planning day — no tasks were moved to Done, so the chart is flat. Work started on Sep 4 and the team completed all 16 points by Sep 7, three days before the Sep 10 deadline. The red line dropping well below the guideline confirms the sprint finished ahead of schedule.

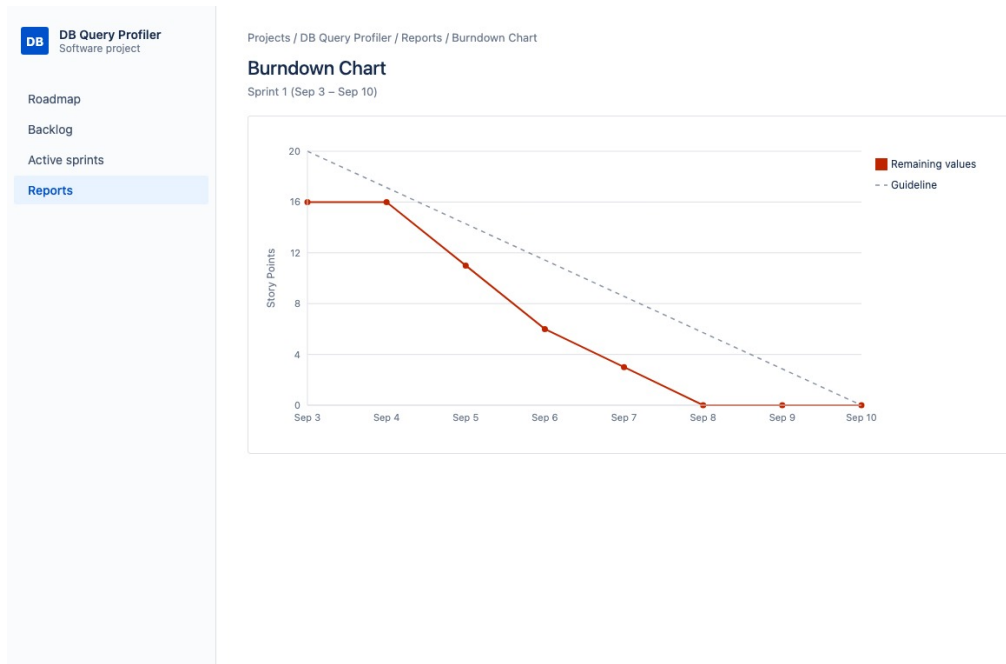


Figure 3 – Burndown Chart for Sprint 1 (Sep 3 – Sep 10)

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6. Sprint 2 – Active Sprint Board (Sep 10 – Sep 17)

Sprint 2 focused on the Index Recommendations, Weekly Digest, and Dashboard epics. All 6 remaining backlog stories were pulled in: DBQP-10(5pts), DBQP-11(3pts), DBQP-12(5pts), DBQP-13(3pts), DBQP-14(5pts), DBQP-15(2pts) — 23 story points in total. The board below was captured mid-sprint: DBQP-10, DBQP-11, and DBQP-13 were done, DBQP-14 was in progress, and DBQP-12 and DBQP-15 were still in To Do.

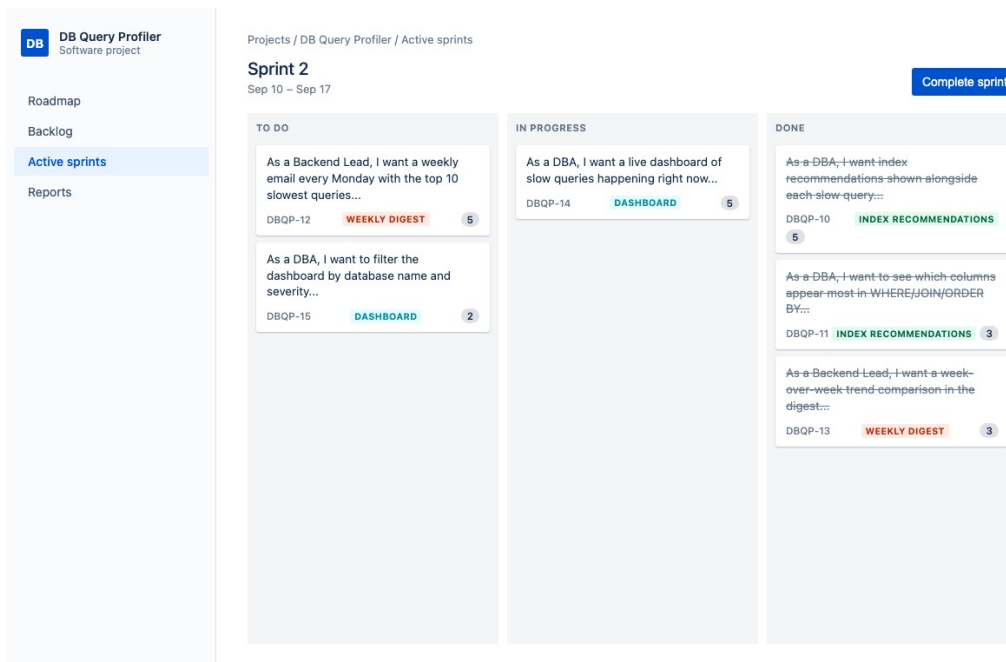


Figure 4 – Sprint 2 active sprint board

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7. Sprint 2 – Burndown Chart

Sprint 2 burndown shows the sprint starting with 23 story points on Sep 10. Sep 11 and Sep 12 saw steady progress (5pts, then 3pts). Sep 13 was a flat day — no stories were closed. Sep 14 picked up again (5pts done), and on Sep 15 the remaining 10 points were all completed, closing the sprint two days before the Sep 17 deadline. The red line finishing well below the guideline shows the team built momentum as the sprint went on.

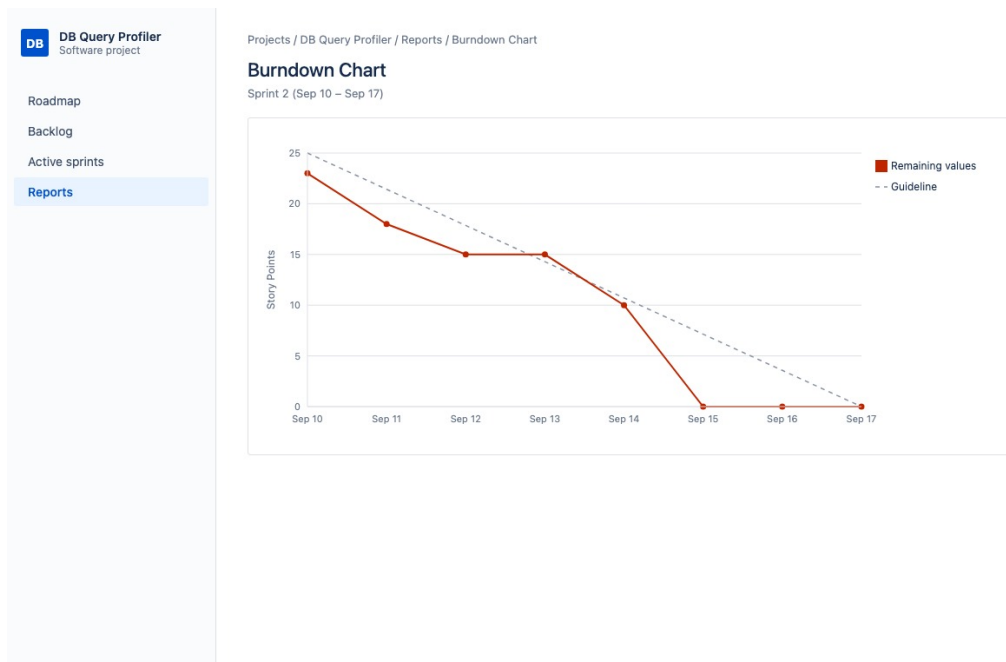


Figure 5 – Burndown Chart for Sprint 2 (Sep 10 – Sep 17)

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8. Reflection

1. Did your estimations reflect the actual effort?

Mostly yes. The story points felt right for most stories. DBQP-7 (live log streaming) was estimated at 5 points and it did turn out to be one of the harder ones to think through. DBQP-15 (dashboard filters) was only 2 points and was actually quite straightforward in Sprint 2. Overall I think the Fibonacci scale helped avoid overthinking the estimates.

2. Was your backlog well-prioritized?

I think so. Putting log ingestion and EXPLAIN plan parsing in Sprint 1 made sense because everything else depends on logs being in the system first. Index recommendations and the dashboard came later in Sprint 2 since they sit on top of the analysis layer. If I were to redo it I would have split Sprint 2 further — 23 points in one sprint is heavy and a real team might have dropped the lower-priority dashboard filter story (DBQP-15) to a Sprint 3.

3. How did your simulated sprint align with your plan?

Sprint 1 had Sep 3 as a planning day with nothing moved yet, but the team got through all 16 points by Sep 7 — three days ahead of schedule. Sprint 2 was heavier at 23 points and had a flat day on Sep 13 where nothing closed, but the team recovered and finished everything by Sep 15, two days early. Both sprints completing before their end dates suggests the story point estimates were slightly conservative.

4. What insights did the burndown chart give about your team's capacity?

The Sprint 1 chart showed we have a slow start pattern — probably spending the first day just reading and understanding the stories. The Sprint 2 chart finishing early suggests the team got faster as they understood the system better. A consistent flat period in the middle of both sprints might indicate a natural dip in productivity mid-week, which is worth keeping in mind for future sprint planning.